SUSTAINABLE CONCRETE

FOR A GREENER FUTURE

High-Performance Concrete Using Supplementary Cementitious Materials and Construction Demolition Waste

Project Presentation

01 | Introduction

Why sustainable concrete matters

- Conventional concrete relies heavily on cement and natural resources.
- Construction and demolition waste creates disposal and pollution challenges.
- This project explores a more sustainable high-performance concrete mix.

02 | Problem Statement

The challenge

- Excessive cement use increases the environmental footprint of concrete.
- Construction demolition waste is often discarded instead of reused.
- A sustainable solution must reduce waste while maintaining concrete performance.

03 | Proposed Solution

Materials used in the project

ALCCOFINE	SILICA FUME	FLY ASH	RECYCLED WASTE
Fine supplementary cementitious material	Helps refine the concrete pore structure	Can partially replace cement	Reuses construction and demolition materials

04 | Working Prototype

Laboratory preparation and testing

- Prepare M70 concrete mixes with selected replacement levels.
- Cast and cure concrete cube and cylinder specimens.
- Test compressive strength and other relevant properties.
- Compare results to identify a suitable sustainable mix.

05 | Benefits & Conclusion

A practical step toward greener construction

- Reduce cement demand and construction waste sent for disposal.
- Conserve natural resources through reuse and material substitution.
- Evaluate strength and durability through laboratory testing.
- Goal: strong, durable concrete with a lower environmental impact.